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IV

PARTIAL DIFFERENTIAL EQUATIONS

- A Partial Differential Equation (P.D.E.) is an eqⁿ which there are 2 or more independent variables and partial differential coefficients w.r.t any of them.

e.g) $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = nu$.

FORMATION OF P.D.E.

I) By eliminating Arbitary Constants:

↳ Given a relation of the form,
$f(x,y,z,a,b) = 0$ where z is a function of x,y and a,b are arbitary constant.

- We diff the given relation w.r.t x & y partially and eliminate the constants a,b to form a P.D.E.

Note:

- If $z = f(x,y)$. Then the following notations are used:

$$p = \frac{\partial z}{\partial x} \; ; \; q = \frac{\partial z}{\partial y} \; , \; r = \frac{\partial^2 z}{\partial x^2} \; ; \; s = \frac{\partial^2 z}{\partial x \partial y} \; ; \; t = \frac{\partial^2 z}{\partial y^2}$$

Q) Form a P.D.E. by eliminating arbitary constants a,b from the following:

i) $2z = \frac{x^2}{a^2} + \frac{y^2}{b^2} \rightarrow (1)$

→ diff - partially w.r.t x

$$2\frac{\partial z}{\partial x} = \frac{2x}{a^2} \Rightarrow \frac{\partial z}{\partial x} = \frac{x}{a^2}$$

$$\Rightarrow \boxed{p = \frac{x}{a^2}} \Rightarrow \boxed{a^2 = \frac{x}{p}}$$

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diff partially w.r.t y

$$2\frac{\partial z}{\partial y} = \frac{2y}{b^2} \Rightarrow \frac{\partial z}{\partial y} = \frac{y}{b^2}$$

$$\Rightarrow q = \frac{y}{b^2} \Rightarrow \boxed{b^2 = \frac{y}{q}}$$

~~diff partially~~ sub in (1).

$$z = \frac{x^2}{\left(\frac{x}{p}\right)} + \frac{y^2}{\left(\frac{y}{q}\right)} \Rightarrow \boxed{z = px + qy}$$

$$\boxed{z = x\frac{\partial z}{\partial x} + y\frac{\partial z}{\partial y}}$$

ii) $z = (x-a)^2 + (y-b)^2 \rightarrow (1)$

→ Diff. partially w.r.t x.

$$\frac{\partial z}{\partial x} = 2(x-a)$$

$$p = 2(x-a)$$

$$\boxed{(x-a) = p/2}$$

Diff. partially w.r.t y.

$$\frac{\partial z}{\partial y} = 2(y-b)$$

$$q = 2(y-b)$$

$$\Rightarrow \boxed{(y-b) = \frac{q}{2}}.$$

sub in (1) $z = \left(\frac{p}{2}\right)^2 + \left(\frac{q}{2}\right)^2$.

$$4z = p^2 + q^2$$

$$4z = \left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2$$

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iii) $z = ax + by + a^2 + b^2$

→ Diff. partially w.r.t x

$$\frac{\partial z}{\partial x} = a$$

$$p = a$$

Diff. partially w.r.t y

$$\frac{\partial z}{\partial y} = b.$$

$$q = b$$

$$\therefore z = x\frac{\partial z}{\partial x} + y\frac{\partial z}{\partial y} + \left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2$$

iv) $z = ax + by + ab$

→ Diff. partially w.r.t x

$$\frac{\partial z}{\partial x} = a$$

$$p = a$$

Diff. partially w.r.t y

$$\frac{\partial z}{\partial y} = b$$

$$q = b.$$

$$z = x\frac{\partial z}{\partial x} + y\frac{\partial z}{\partial y} + \frac{\partial^2 z}{\partial x \partial y}$$

v) $z = ax^2 + by^2.$

→ Diff. partially w.r.t x

$$\frac{\partial z}{\partial x} = 2ax$$

$$p = 2ax$$

$$\Rightarrow \boxed{a = \frac{p}{2x}}$$

$$a = \frac{1}{2x}\frac{\partial z}{\partial x}$$

Diff partially w.r.t

$$\frac{\partial z}{\partial y} = 2by$$

$$q = 2by$$

$$\boxed{b = \frac{q}{2y}}$$

$$b = \frac{1}{2y}\frac{\partial z}{\partial y}$$

$$z = \frac{1}{2}\left[x\frac{\partial z}{\partial x} + y\frac{\partial z}{\partial y}\right]$$

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ii) $z = a \log\left(\frac{b(y-1)}{1-x}\right)$.

→ $\frac{\partial z}{\partial x} = -a\, b(y-1).(1-x)$

$p = -ab(y-1)(1-x)$ ⇒

Diff. partially w.r.t y

$\frac{\partial z}{\partial y} = \frac{ab}{(1-x)} \cdot \frac{1}{(y-1)}$

$\frac{\partial z}{\partial y} = \frac{ab}{(1-x)(y-1)}$

$q = \frac{ab}{(1-x)(y-1)}$

$z = a\left[\log(b(y-1)) - \log(1-x)\right]$

$z = a \log(b(y-1)) - a \log(1-x)$

$z = a\log b + a\log(y-1) - a\log(1-x)$

$\frac{\partial z}{\partial x} = \frac{a}{1-x}$ $\qquad$ $\frac{\partial z}{\partial y} = \frac{a}{(y-1)}$

$a = p(1-x)$ $\qquad$ $\frac{\partial z}{\partial y} = \frac{a}{y-1}$

$q(y-1) = a$

⇒ $p(1-x) = q(y-1)$

⇒ $p - px = qy - q$

⇒ $p + q - px - qy = 0$ ⇒ $p(1-x) + q(1-y) = 0$

$\frac{\partial z}{\partial x}(1-x) + \frac{\partial z}{\partial y}(1-y) = 0$ //.

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II) By eliminating arbitrary functions

* Suppose 'z' is a function of one arbitrary function. we differentiate the given relation w.r.t. x and y and eliminate the arbitrary function.
* Suppose 'z' is a function of two arbitrary functions. we find partial derivatives upto second order and use the necessary partial derivative to form the P.D.E. by eliminating the arbitrary functions.

a) Form P.D.E. by eliminating arbitrary function.

i) $z = f(x^2 - y^2)$

$\rightarrow z = f(x^2 - y^2) \rightarrow (1)$

Diff. (1) partially w.r.t x

$$\frac{\partial z}{\partial x} = f'(x^2 - y^2) \cdot 2x$$

$$p = 2x f'(x^2 - y^2) \Rightarrow \boxed{f'(x^2 - y^2) = p/2x} \rightarrow (2).$$

Diff. (1) partially w.r.t. y

$$\frac{\partial z}{\partial y} = f'(x^2 - y^2)(-2y)$$

$$q = -2y f'(x^2 - y^2) \Rightarrow \boxed{f'(x^2 - y^2) = -q/2y} \rightarrow (3)$$

eq (2) = eq (3).

$$\frac{p}{2x} = \frac{-q}{2y} \Rightarrow py = -qx$$

$$\Rightarrow \Rightarrow y\frac{\partial z}{\partial x} + x\frac{\partial z}{\partial y} = 0 //$$

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ii) $z = f(x) + e^y g(x)$

→ $z = f(x) + e^y g(x) \rightarrow (1)$.

Since z is made up of two arbitrary function we find partial derivatives upto second order.

$$p = \frac{\partial z}{\partial x} = f'(x) + e^y g'(x)$$

$$r = \frac{\partial^2 z}{\partial x^2} = f''(x) + e^y g''(x)$$

$$q = \frac{\partial z}{\partial y} = e^y g(x)$$

$$t = \frac{\partial^2 z}{\partial y^2} = e^y g(x)$$

$$s = \frac{\partial^2 z}{\partial x \partial y} = \frac{\partial}{\partial x}\left(\frac{\partial z}{\partial y}\right) = \frac{\partial}{\partial x}\left(e^y g(x)\right) = e^y g'(x)$$

Hence we get

$q = t$

$$\frac{\partial z}{\partial y} = \frac{\partial^2 z}{\partial y^2}$$

iii) $z = f(x-y) + g(x+y)$

→ $z = f(x-y) + g(x+y) \rightarrow (1)$

Diff. partially w.r.t x

$$p = \frac{\partial z}{\partial x} = f'(x-y) + g'(x+y)$$

$$q = \frac{\partial z}{\partial y} = -f'(x-y) + g'(x+y)$$

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$$r = \frac{\partial^2 z}{\partial x^2} = f''(x-y) + g''(x+y)$$

$$s = \frac{\partial^2 z}{\partial x \partial y} = -f''(x-y) + g''(x+y)$$

$$t = \frac{\partial^2 z}{\partial y^2} = f''(x-y) + g''(x+y)$$

$$r = t$$

$$\frac{\partial^2 z}{\partial x^2} = \frac{\partial^2 z}{\partial y^2}$$

iv) $z = f(x+iy) + g(x-iy)$

→ $z = f(x+iy) + g(x-iy)$

$$p = \frac{\partial z}{\partial x} = f'(x+iy) + g'(x-iy)$$

$$q = \frac{\partial z}{\partial y} = if'(x+iy) + i(g'(x-iy))$$

$$= i\left[f'(x+iy) - g'(x-iy)\right]$$

$$r = \frac{\partial^2 z}{\partial x^2} = f''(x+iy) + g''(x-iy)$$

$$s = \frac{\partial^2 z}{\partial x \partial y} = \left(f''(x+iy) - g''(x-iy)\right) = -\left(f''(x+iy) + g''(x-iy)\right)$$

$$t = \frac{\partial^2 z}{\partial y^2} = -f''(x+iy) + g''(x-iy)$$

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v) $z = y^2 + 2f\left(\frac{1}{x} + \log y\right)$

$\rightarrow p = 2f'\left(\frac{1}{x} + \log y\right)\cdot\frac{-1}{x^2}$

$q = 2y + 2f'\left(\frac{1}{x} + \log y\right)\cdot\frac{1}{y}$

$\Rightarrow px^2 = -2f'\left(\frac{1}{x} + \log y\right)$

$yq - 2y^2 = 2f'\left(\frac{1}{x} + \log y\right)$

$\Rightarrow -px^2 = qy - 2y^2$

$\therefore -x^2\frac{\partial z}{\partial x} - y\frac{\partial z}{\partial y} + 2y^2 = 0$

vi) $z = y(f(x)) + x(g(y))$

$\rightarrow$

vii) $x+y+z = f(x^2+y^2+z^2)$

→ $p = \frac{\partial z}{\partial x}$

diff partially w.r.t x

$$1+\frac{\partial z}{\partial x} = f'(x^2+y^2+z^2)\cdot\left(2x+2z\frac{\partial z}{\partial x}\right)$$

$$1+p = 2f'(x^2+y^2+z^2)(x+pz)$$

$$\Rightarrow \boxed{\frac{1+p}{2(x+pz)} = 2f'(x^2+y^2+z^2)}$$

diff partially w.r.t y.

$$1+\frac{\partial z}{\partial y} = f'(x^2+y^2+z^2)\left(2y+2z\frac{\partial z}{\partial y}\right)$$

$$1+q = 2f'(x^2+y^2+z^2)(y+qz)$$

$$\Rightarrow \boxed{\frac{1+q}{2(y+qz)} = 2f'(x^2+y^2+z^2)}$$

$$\Rightarrow \frac{1+p}{x+pz} = \frac{1+q}{y+qz}$$

$$\Rightarrow y+yp+qz+pqz = x+xq+pz+pqz$$

$$\Rightarrow y-x+yp+qz-pz+xq = 0$$

$$y-x+y\frac{\partial z}{\partial x}-x\frac{\partial z}{\partial y}+z\left(\frac{\partial z}{\partial y}-\frac{\partial z}{\partial x}\right)=0$$

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viii) $xyz = f(x+y+z)$

→ diff partially w.r.t x

$$y\left(z + x\frac{\partial z}{\partial x}\right) = f'(x+y+z)\cdot\left(1+\frac{\partial z}{\partial x}\right)$$

$$\Rightarrow f'(x+y+z) = \frac{y(z+xp)}{(1+p)} \rightarrow (1)$$

diff partially w.r.t y

$$x\left(z + y\frac{\partial z}{\partial y}\right) = f'(x+y+z)\left(1+\frac{\partial z}{\partial y}\right)$$

$$\Rightarrow f'(x+y+z) = \frac{x(z+yq)}{(1+q)} \rightarrow (2)$$

eq (1) = eq (2)

$$\frac{y(z+xp)}{1+p} = \frac{x(z+yq)}{1+q}$$

$$\Rightarrow (y+qy)(z+xp) = (x+xp)(z+yq)$$

$$\Rightarrow yz + xyp + qyz + xpqy = xz + xyq + xpz + xpyq$$

$$\Rightarrow yz + xyp + yzq = xz + xyq + xzp$$

$$\Rightarrow yz - xz + xy\left(\frac{\partial z}{\partial x} - \frac{\partial z}{\partial y}\right) + z\left(y\frac{\partial z}{\partial y} - z\frac{\partial z}{\partial x}\right) = 0$$

Note:

1. If expression is of the form $f(u,v)=0$ where u,v are some functions of x,y,z, z be dependent variable. We use chain rule to form the P.D.E.

$F \rightarrow (u,v) \rightarrow (x,y)$

Q) Solve the following P.D.E.

i) $F(xy+z^2,\ x+y+z)=0$

→

$u = xy+z^2 \qquad v = x+y+z$

$$\frac{\partial f}{\partial x} = \frac{\partial f}{\partial u}\times\frac{\partial u}{\partial x} + \frac{\partial f}{\partial v}\times\frac{\partial v}{\partial x} = 0 \qquad f\to(u,v)\to(x,y,z)$$

$$\frac{\partial f}{\partial u}\times\left(y+2z\frac{\partial z}{\partial x}\right)+\frac{\partial f}{\partial v}\left(1+\frac{\partial z}{\partial x}\right)=0$$

$$\Rightarrow \frac{\partial f}{\partial u}\left(y+2zp\right) = -\frac{\partial f}{\partial v}\left(1+p\right) \rightarrow (1)$$

$$\frac{\partial f}{\partial y} = \frac{\partial f}{\partial u}\times\frac{\partial u}{\partial y}+\frac{\partial f}{\partial v}\times\frac{\partial v}{\partial y}=0$$

$$\frac{\partial f}{\partial u}\left(x+2z\frac{\partial z}{\partial y}\right)+\frac{\partial f}{\partial v}\left(1+\frac{\partial z}{\partial x}\right)=0$$

$$\Rightarrow \frac{\partial f}{\partial u}\left(x+2zq\right) = -\frac{\partial f}{\partial v}\left(1+q\right) \rightarrow (2)$$

$1 \div 2$

$$\frac{y+2zp}{x+2zq} = \frac{1+p}{1+q}$$

$$(1+q)(y+2zp) = (1+p)(x+2zq)$$

$$\Rightarrow y+2zp+yq+2zpq = x+2zq+xp+2zpq$$

$$(x-y)+2z(q-p)+xp-yq=0$$

$$\Rightarrow (x-y)+2z\left(\frac{\partial z}{\partial y}-\frac{\partial z}{\partial x}\right)+x\frac{\partial z}{\partial x}-y\frac{\partial z}{\partial y}=0$$

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ii) $F(x^2+y^2, z-xy) = 0$

→

$u = x^2+y^2 \qquad v = z-xy$

$$\frac{\partial F}{\partial x} = \frac{\partial f}{\partial u} \times \frac{\partial u}{\partial x} + \frac{\partial f}{\partial v} \times \frac{\partial v}{\partial x} = 0$$

$$\frac{\partial F}{\partial u}(2x) = -\frac{\partial f}{\partial v}\left(\frac{\partial z}{\partial x} - y\right)$$

$$2x\frac{\partial F}{\partial u} = -\frac{\partial f}{\partial v}(p-y) \rightarrow (1).$$

$$\frac{\partial F}{\partial y} = \frac{\partial F}{\partial u}\left(\frac{\partial u}{\partial y}\right) + \frac{\partial f}{\partial v} \times \frac{\partial v}{\partial y} = 0$$

$$= \frac{\partial f}{\partial u}(2y) = -\frac{\partial f}{\partial v}\left(\frac{\partial z}{\partial y} - x\right)$$

$$2y\frac{\partial f}{\partial u} = -\frac{\partial f}{\partial v}(q-x) \rightarrow (2).$$

(1) ÷ (2)

$$\frac{x}{y} = \frac{p-y}{q-x}$$

$$\Rightarrow xq - x^2 = yp - y^2$$

$$\Rightarrow x^2 - y^2 + yp - xq$$

$$\Rightarrow x^2 - y^2 + y\left(\frac{\partial z}{\partial x}\right) - x\left(\frac{\partial z}{\partial y}\right)$$

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iii) $f(x^2+ay z+y^2+axz)=0$ $\quad f\to(u,v)\to(x,y)$

$\rightarrow$

$$\frac{\partial f}{\partial x}=\frac{\partial f}{\partial u}\times\frac{\partial u}{\partial x}+\frac{\partial f}{\partial v}\times\frac{\partial v}{\partial x}=0$$

$$=\frac{\partial f}{\partial u}\left(2x+2y\frac{\partial z}{\partial x}\right)+\frac{\partial f}{\partial v}\left(2x\frac{\partial z}{\partial x}+2z\right)=0$$

$$\Rightarrow \frac{\partial f}{\partial u}\left(x+yp\right)=-\frac{\partial f}{\partial v}\left(xp+z\right)\rightarrow(1)$$

$$\frac{\partial f}{\partial y}=\frac{\partial f}{\partial u}\times\frac{\partial u}{\partial y}+\frac{\partial f}{\partial v}\times\frac{\partial v}{\partial y}=0$$

$$=\frac{\partial f}{\partial u}\left(2y\frac{\partial z}{\partial y}+2z\right)+\frac{\partial f}{\partial v}\left(2y+2x\frac{\partial z}{\partial y}\right)$$

$$\Rightarrow \frac{\partial f}{\partial u}\left(yq+z\right)=-\frac{\partial f}{\partial v}\left(y+xq\right)\rightarrow(2)$$

$\Rightarrow$ (1) ÷ (2)

$$\frac{x+yp}{yq+z}=\frac{xp+z}{y+xq}$$

$$\Rightarrow xy+x^2q+y^2p+xypq=xypq+yqz+xzp+z^2$$

$$xy+x^2\frac{\partial z}{\partial y}+y^2\frac{\partial z}{\partial x}-yz\frac{\partial z}{\partial y}+xz\frac{\partial z}{\partial x}+z^2$$

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SOLUTION OF A PARTIAL D.E.

Solution of a P.D.E. is a relation b/w the dependent and independent variable satisfying the equation.

Eg) The general soln of $dz = px + qy$ is $z = ax^2 + by^2$ and particular soln is $z = 2x^2 - y^2$

General solution $f(x, y, z, a, b) = 0$ of a first order P.D.E. which contains 2 arbitrary constants is called a complete interval (General soln)

Soln obtained from complete interval by assigning particular values to arbitrary constants is called particular solution.

SOLUTION OF P.D.E. BY METHOD OF DIRECT INTEGRATION

Q) Solve the following D.E. by method of Direct Integration.

i) $\frac{\partial^3 z}{\partial x^2 \partial y} + 18xy^2 + 2\sin(2x - y) = 0$

→ Integrating w.r.t. x

$$\int \frac{\partial^3 z}{\partial x^2 \partial y}\, dx + \int 18xy^2\, dx + \int 2\sin(2x - y) = 0$$

$$\Rightarrow \frac{\partial^2 z}{\partial x \partial y} + 9x^2y^2 + \frac{\cos(2x - y)}{2} = f(y)$$

Integrating w.r.t x

$$\frac{\partial z}{\partial y} + 3x^3y^2 - \frac{\sin(2x - y)}{4} = g(y).x + h(y)$$

Integrating w.r.t y

$$z + x^3y^3 - \frac{\cos(2x-y)}{4} = x\int g(y) + \int h(y) + f(x)$$

$$\therefore z = \frac{\cos(2x-y)}{4} - x^3y^3 + x\,G(y) + H(y) + f(x)$$

ii) $\frac{\partial^2 z}{\partial x \partial y} = \frac{x}{y} + a$

→ Integrating w.r.t x

$$\int \frac{\partial^2 z}{\partial x \partial y}\,dx = \int \frac{x}{y}\,dx + \int a\,dx$$

$$\frac{\partial z}{\partial y} = \frac{x^2}{2y} + ax + g(y)$$

Integrating w.r.t y.

$$\int \frac{\partial^2 z}{\partial x \partial y}\,dy = \int \frac{x^2}{2y}\,dy + \int ax\,dy + \int g(y)\,dy$$

$$\therefore z = \frac{x^2}{2}\log|y| + axy + G(y) + h(x)$$

iii) $\frac{\partial^3 z}{\partial x^2 \partial y} = \cos(2x+3y)$

→ Integrating w.r.t x

$$\int \frac{\partial^3 z}{\partial x^2 \partial y}\,dx = \int \cos(2x+3y)\,dx$$

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$$\frac{\partial^2 z}{\partial x \partial y} = -\frac{\sin(2x+3y)}{2} + g(y)$$

Integrating w.r.t x

$$\frac{\partial z}{\partial y} = -\frac{\cos(2x+3y)}{4} + x\,g(y) + h(y)$$

Integrating w.r.t y.

$$z = -\frac{\sin(2x+3y)}{12} + x\,G(y) + H(y) + f(x)$$

(iv) $\frac{\partial^2 z}{\partial x^2} + z = 0$ when $x=0 \Rightarrow z = e^y$ & $\frac{\partial z}{\partial x} = 1$

→ Given $\frac{\partial^2 z}{\partial x^2} + z = 0$

Treating this as second order O.D.E. in x (taking y as constant)

This can be re-written as

$$\frac{d^2 z}{dx^2} + z = 0 \qquad (D^2+1)z = 0$$

A.E.: $m^2+1=0$

$m^2 = -1$

$m = \pm i$. $\quad \alpha = 0 \quad \beta = 1$

$z = C_1 \cos x + C_2 \sin x$

$\Rightarrow z = g(y)\cos x + h(y)\sin x$

When $x=0 \Rightarrow z = e^y$

$e^y = g(y)$

$\Rightarrow z = e^y \cos x + \sin x$

$z = e^y \cos x + C_2 \sin x$

$\frac{\partial z}{\partial x} = 1 = -e^y \sin x + C_2 \cos x$

$1 = C_2(y)$